UNIT 1 · TOPIC 1.2

Suspicious Website Logins

Assignment

Instructions: 15 practice questions + 3 coding assignment(s), 30 points total. Show your work for short-answer items. Coding assignments may be completed here or in the interactive auto-graded environment on the course site (same problem, same points).

Name: **Date:** **Class:**

Practice Questions

1.2.a · Password-attack signs (failed attempts, unusual times, unknown devices)

1. An authentication log shows 47 failed logins for the same account over 90 seconds, then a success. What type of attack does this pattern indicate? (1 pt)

- ○ A. Phishing
- ○ B. Brute-force / dictionary attack
- ○ C. SQL injection
- ○ D. Man-in-the-middle

2. A defender sets an alert for '5 failed logins within 60 seconds for the same user.' Why is the TIME WINDOW included in the rule, not just the raw count? (1 pt)

- ○ A. To distinguish slow manual mistakes from fast automated guessing, reducing false positives
- ○ B. So the alert ignores failures that come from the user's usual device
- ○ C. Because logs only keep the last 60 seconds of events
- ○ D. So that attackers who guess slowly are also caught

3. Which of the following are recognized PASSWORD-ATTACK SIGNALS in an authentication log? Select ALL that apply. (3 pts)

- □ A. Many failed logins for one account in a short time
- □ B. A login from a device never seen before
- □ C. A successful login at 3 a.m. when the user normally works 9-5
- □ D. A user changing their display name in their profile settings

(Select ALL that apply.)

4. Credential stuffing differs from a classic dictionary attack because: (1 pt)

- ○ A. It tries every possible character combination
- ○ B. It uses real username/password pairs stolen from a different service's breach
- ○ C. It requires physical access to the target device
- ○ D. It only targets administrator accounts

5. An account lockout fires AFTER five failures. A defender notices the log has a SUCCESS before the lockout triggered. What should they conclude? (1 pt)

1.2.b · Weak authentication patterns; adversary dictionary construction

6. An attacker has a 14-million-entry wordlist from a known breach. What type of attack does this enable? (1 pt)

- ○ A. Brute-force (exhaustive)
- ○ B. SQL injection
- ○ C. Replay attack
- ○ D. Dictionary attack

7. Which of the following make a password MORE vulnerable to a dictionary attack? Select ALL that apply. (3 pts)

- □ A. Using a common word as the base
- □ B. Substituting 'a' with '@'
- □ C. Length of 20+ random characters
- □ D. Appending '1' to a common word

(Select ALL that apply.)

8. Under NIST SP 800-63B-4 (2025), the minimum length for a password used on its own, without MFA, is: (1 pt)

- ○ A. 6 characters
- ○ B. 8 characters
- ○ C. 12 characters
- ○ D. 15 characters

9. A password 'HelloWorld111' passes a complexity check (has uppercase, lowercase, digits). Which weakness flag does the automated checker still raise? (1 pt)

- ○ A. too_short
- ○ B. in_dictionary
- ○ C. sequential_or_repeated
- ○ D. only_lowercase

10. Name the technique attackers use to automatically generate hundreds of variants from a single wordlist entry (e.g., 'password' -> 'P@ssw0rd', 'password1', 'PASSWORD'). (1 pt)

1.2.c · Strengthening authentication: length/complexity, randomness, MFA

11. According to NIST SP 800-63B, what is the primary driver of password strength? (1 pt)

- ○ A. Mandatory symbol requirements
- ○ B. Mandatory uppercase + digit rules
- ○ C. Frequency of rotation
- ○ D. Password length

12. Which of the following are factors in a TOTP (Time-based One-Time Password) computation? Select ALL that apply. (3 pts)

- □ A. A shared secret key K
- □ B. The current Unix timestamp

□ C. The user's password

□ D. A 30-second time step window

(Select ALL that apply.)

13. An attacker intercepts a valid TOTP code and tries to use it 45 seconds later. What happens? (1 pt)

○ A. The code still works because TOTP has a 60-second window

○ B. The server cannot tell the difference

○ C. The code works once more then expires

○ D. The code is rejected because the time step has changed

14. Name the two factors that MFA combines to protect a Wi-Fi login. Use the 'something you know / something you have' framework. (1 pt)

15. In the TOTP algorithm, what is the 'time step' T for a Unix timestamp of 1700000075? (1 pt)

○ A. 1700000075

○ B. 30

○ C. 56666002

○ D. 56666669

Coding Assignments (Python 3)

1.2.a · Password-attack signs (failed attempts, unusual times, unknown devices)

3 pts · auto-graded on the site

Coding problem 1.2.a: Detect brute-force login attempts

Skill: Detect Attacks · **Maps to:** 1.2.A · **Language:** Python 3 (pytest)

Problem

Build the detector a login-monitoring system would use. Given a list of authentication events, return every username that has five or more failed login attempts within any 60-second window.

This is the core algorithm behind real intrusion-detection rules (e.g., fail2ban's default policy). The defender's goal is to catch automated password-guessing tools before they succeed.

Function contract

Implement `solve(events: list[dict]) -> list[str]` in `starter.py` (see `interface.md` / `blackbox.yaml`). Return the sorted, de-duplicated list of usernames flagged as brute-force targets.

Input

A list of event dicts. Each dict has:

- `ts` (`int`), Unix timestamp in seconds of the login attempt
- `user` (`str`), the username targeted
- ok (`bool`), `True` if the login succeeded, `False` if it failed

Events may arrive in any order (do not assume sorted by timestamp).

Output

A sorted `list[str]` of unique usernames that triggered the brute-force threshold: 5 or more **failed** logins (where `ok == False`) within any 60-second window. A window is valid when `max_ts - min_ts < 60` (strict less-than, so exactly 60 is NOT within the window). Successes are ignored for counting purposes.

Examples

```
events = [
    {"ts": 1000, "user": "alice", "ok": False},
    {"ts": 1010, "user": "alice", "ok": False},
    {"ts": 1020, "user": "alice", "ok": False},
    {"ts": 1030, "user": "alice", "ok": False},
    {"ts": 1040, "user": "alice", "ok": False},  # 5 failures in 40s -> flagged
    {"ts": 1000, "user": "bob",   "ok": False},
    {"ts": 1100, "user": "bob",   "ok": False},  # 2 failures, not enough -> not flagged
]
solve(events) -> ["alice"]
```

```
solve([]) -> []
```

```
# Four failures in window: below threshold
events = [{"ts": i*10, "user": "carol", "ok": False} for i in range(4)]
solve(events) -> []
```

Constraints

- Deterministic, offline, standard library only.
- Events may be unsorted; your solution must handle any ordering.
- The 60-second window: at least 5 failures where `max_ts - min_ts < 60`.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 1.2.a. Student completes the TODO.
This file intentionally FAILS the tests until implemented."""
from typing import List, Dict, Any


def solve(events: List[Dict[str, Any]]) -> List[str]:
    """Return sorted, de-duplicated usernames with >=5 failed logins in any 60s window.

    Args:
        events: list of dicts, each with keys:
                  ts   (int)  - Unix timestamp in seconds
                  user (str)  - username targeted
                  ok   (bool) - True=success, False=failure
    Returns:
        Sorted list of usernames flagged as brute-force targets.
        See prompt.md and interface.md for the exact detection rule.
    """
    # TODO: for each user, collect failure timestamps, sort them, then use a
    #       sliding window of size 5 to check if ts[i+4] - ts[i] < 60.
```

```
    #       Return a sorted list of flagged usernames.
    raise NotImplementedError("implement solve() — see prompt.md")
```

1.2.b · Weak authentication patterns; adversary dictionary construction

3 pts · auto-graded on the site

Coding problem 1.2.b: Password weakness reporter

Skill: Analyze Risk · **Maps to:** 1.2.B · **Language:** Python 3 (pytest)

Problem

Build the password-weakness checker an authentication backend would use to give users actionable feedback when they choose a new password. Given a password string, return a report dict identifying whether the password is weak and which specific rules it violates.

Function contract

Implement `solve(password: str) -> dict` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Input

A single string, the candidate password. May be empty.

Output

A dict with exactly two keys:

- `"weak"` (`bool`), `True` if the password fails ANY rule; `False` only if it passes all rules.
- `"reasons"` (`list[str]`), sorted list of rule-labels for each rule the password fails. Empty list if the password passes all rules.

Possible reason labels (exact strings, lowercase):

- `"too_short"`: length < 12 characters
- `"in_dictionary"`: password (lowercased) is in the built-in common wordlist
- `"only_lowercase"`: contains no uppercase letters, digits, or non-alphabetic characters
- `"sequential_or_repeated"`: contains a sequential run of 3+ characters (e.g. `abc`, `123`, `xyz`) OR a repeated run of 3+ identical characters (e.g. `aaa`, `111`)

Examples

```
solve("password")
-> {"weak": True, "reasons": ["in_dictionary", "only_lowercase", "too_short"]}

solve("P@ssword!2024")
-> {"weak": True, "reasons": ["in_dictionary"]}

solve("Tr0ub4dor&3abcde")
-> {"weak": True, "reasons": ["sequential_or_repeated"]}

solve("mXk9#vLq2@nP")
-> {"weak": False, "reasons": []}
```

Constraints

- Deterministic, offline, standard library only.

- The common wordlist is defined in `interface.md` and must be embedded in your solution.
- Reasons list is always sorted ascending.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 1.2.b. Student completes the TODO.
This file intentionally FAILS the tests until implemented."""


def solve(password: str) -> dict:
    """Return a weakness report dict for the given password.

    Args:
        password: the candidate password string (may be empty).
    Returns:
        {"weak": bool, "reasons": list[str]} where reasons is a sorted list of
        triggered rule labels. Possible labels:
          "too_short"            -- len(password) < 12
          "in_dictionary"        -- password.lower() is in the common wordlist
          "only_lowercase"       -- password contains only a-z characters
          "sequential_or_repeated" -- run of 3+ sequential or identical chars
        See interface.md for the exact COMMON_WORDS set and detection algorithm.
    """
    # TODO: implement all four rules and return {"weak": ..., "reasons": [...]}
    raise NotImplementedError("implement solve() -- see prompt.md")
```

1.2.c · Strengthening authentication: length/complexity, randomness, MFA

3 pts · auto-graded on the site

Coding problem -- 1.2.c: TOTP Verifier

Skill: Mitigate Risk · **Maps to:** 1.2.C · **Language:** Python 3 (pytest)

Problem

Implement the server-side Time-based One-Time Password (TOTP) verifier that an authentication backend would call to check a user's second factor. Given the shared secret, a submitted code, and the current time, return True if the code is valid for the current 30-second window.

This is the algorithm defined in RFC 6238 (TOTP) / RFC 4226 (HOTP), implemented using only Python's standard library `hmac` and `hashlib` modules.

Function contract

Implement `solve(secret: str, code: str, t: int) -> bool` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Input

- `secret` (`str`) -- the shared secret key, encoded as an **even-length lowercase hex string** (e.g. `"0a1b2c3d"` represents 4 bytes: `b'\x0a\x1b\x2c\x3d'`).
- `code` (`str`) -- the 6-digit code submitted by the user, zero-padded if necessary (e.g. `"007342"` or `"123456"`).

- `t` (`int`) -- the current Unix timestamp (seconds since epoch) at the moment of verification.

Output

`True` if the computed TOTP code for the current 30-second step matches `code`; `False` otherwise.

Algorithm (exact steps, deterministic)

1. Decode the secret: `key = bytes.fromhex(secret)` 2. Compute the time step: `step = t // 30` 3. Encode the step as an 8-byte big-endian integer: `msg = step.to_bytes(8, 'big')` 4. Compute HMAC-SHA1: `digest = hmac.new(key, msg, hashlib.sha1).digest()` (This produces a 20-byte bytes object.) 5. Dynamic truncation: a. `offset = digest[19] & 0x0f` b. `p = ((digest[offset] & 0x7f) << 24) | (digest[offset+1] << 16) | (digest[offset+2] << 8) | digest[offset+3]` 6. Compute the OTP: `otp = p % 1_000_000` 7. Zero-pad: `expected = str(otp).zfill(6)` 8. Return `code == expected`

Examples

```
# Using key=bytes.fromhex("31323334353637383930313233343536373839 30") (RFC 4226 test vector)
# At t=59 (step=1): expected code is "287082"
solve("31323334353637383930313233343536373839 30", "287082", 59)   -> True
solve("31323334353637383930313233343536373839 30", "287082", 60)   -> False  # step changed
solve("31323334353637383930313233343536373839 30", "000000", 59)   -> False  # wrong code
```

Constraints

- Deterministic, offline, stdlib only (`hmac`, `hashlib`).
- Accept only the CURRENT step (no window drift / no adjacent steps).
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 1.2.c. Student completes the TODO.
This file intentionally FAILS the tests until implemented."""


def solve(secret: str, code: str, t: int) -> bool:
    """Verify a 6-digit TOTP code against the shared secret at Unix timestamp t.

    Args:
        secret: even-length hex string encoding the shared key (e.g. "0a1b2c3d").
        code:   6-character zero-padded decimal string (e.g. "007342").
        t:      current Unix timestamp (int, seconds since epoch).
    Returns:
        True if code matches TOTP(secret, t // 30); False otherwise.

    Algorithm (from interface.md):
      1. key    = bytes.fromhex(secret)
      2. step   = t // 30
      3. msg    = step.to_bytes(8, byteorder='big')
      4. digest = hmac.new(key, msg, hashlib.sha1).digest()   # 20 bytes
      5. offset = digest[19] & 0x0f
      6. p      = ((digest[offset] & 0x7f) << 24) | (digest[offset+1] << 16)
                  | (digest[offset+2] << 8) | digest[offset+3]
      7. otp    = p % 1_000_000
      8. return code == str(otp).zfill(6)
```

```
Import hmac and hashlib from the standard library only.
"""
# TODO: implement the TOTP verifier -- see prompt.md and interface.md
raise NotImplementedError("implement solve() -- see prompt.md")
```